Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE

GRADE 10

NOVEMBER 2017

MATHEMATICAL LITERACY P2

MARKS: **75**

TIME: **1½ hours**

XMLITE2

This question paper consists of 6 pages and an ADDENDUM with 6 ANNEXURES.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ANNEXURES in the ADDENDUM to answer the following questions:

   ANNEXURE A for QUESTION 1.1
   ANNEXURE B for QUESTION 2.1
   ANNEXURE C for QUESTION 2.2
   ANNEXURE D for QUESTION 3.1
   ANNEXURE E for QUESTION 4.1
   ANNEXURE F for QUESTION 4.2

3. Number the answers correctly according to the numbering system used in this question paper.

4. Start EACH question on a NEW page.

5. You may use an approved calculator (non-programmable and non-graphical), unless stated otherwise.

6. Show ALL calculations clearly.

7. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

8. Indicate units of measurement, where applicable.

9. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

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QUESTION 1

1.1 Mr. Buqa has an account with Exclusive Clothing Store and receive an account statement every month. ANNEXURE A shows one of Mr. Buqa's account statements.

Use the account statement in ANNEXURE A to answer the questions that follow.

1.1.1 Show with the necessary calculations how the purchases for the month has been calculated. (2)

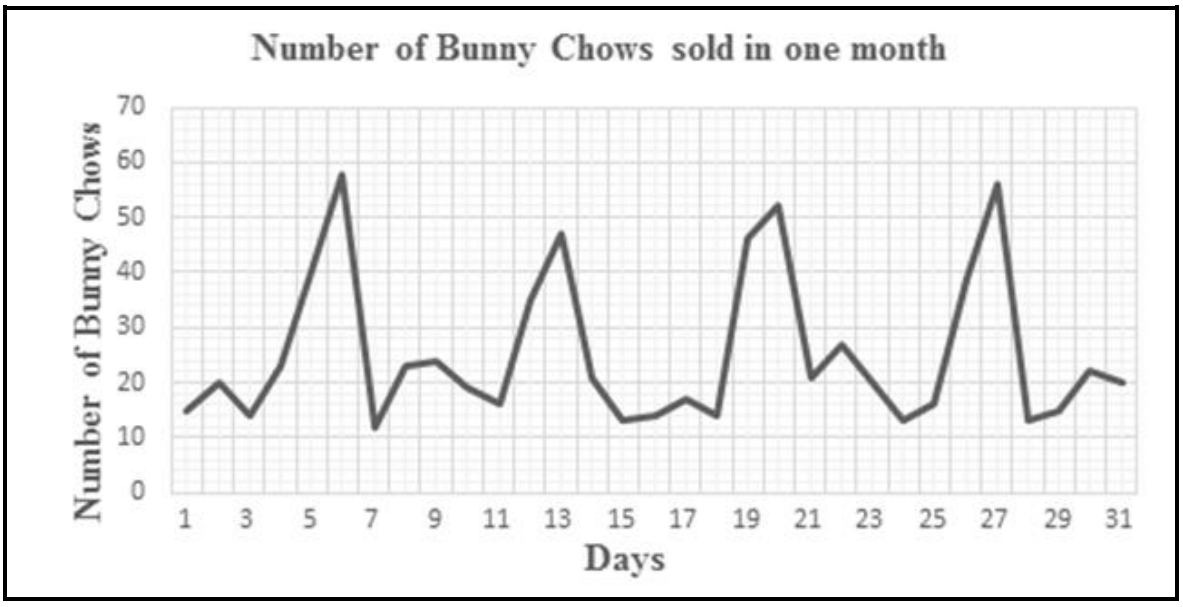
1.1.2 Determine the missing value, A (interest on the money outstanding). (4)

1.1.3 Exclusive Clothing Store calculates the amount owing using a percentage. Calculate the percentage they have used to determine the amount that Mr. Buqa has to pay on the balance that he owes. (3)

1.1.4 Give ONE reason why Mr. Buqa returned the dress. (2)

1.1.5 Mr. Buqa stated that it is unfair of the store not refund him for the underwear he bought, because it will be of no use to his children. Give ONE reason to support the store for not taking back the underwear. (2)

1.2 Mrs. Buqa sells "Bunny Chows" at her house to support her husband with the financial running of the household. Below is a graph representing her sales for one month.



Note: A Bunny Chow is a quarter loaf of bread where the soft part of the bread is removed. The inside is then filled with various fillings and the softer part is put back on top.

Refer to the graph above to answer the following questions:

1.2.1 Explain, with justification, whether the given data is discrete or continuous. (3)

1.2.2 Mrs. Buqa states that she sells more Bunny Chows on weekends than on weekdays. Can it be said with certainty that her statement is true? Give a reason for your answer. (2)

1.2.3 Describe the possible trend of the sales over the last four days of the month. (3)

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QUESTION 2

2.1 Mary-Ann has a Gold Current Account at Absa Bank. The fee structure of Absa Bank is given in ANNEXURE B.

Use the fee structure in ANNEXURE B to answer the questions below.

2.1.1 Calculate the difference in price between 2016 and 2017 to deposit R500 at an Automatic Teller Machine (ATM). (4)

2.1.2 Mary-Ann stated that the percentage change for a withdrawal from ABSA increased by fifty percent more than the withdrawal from another ATM. Show, using calculations, whether her statement is valid or not. (5)

2.2 Mary-Ann received a basic bread recipe from a friend in America. The basic bread recipe is given in ANNEXURE C.

You may use the following conversions:
1 ounce = 28 grams
1 cup = 250 mℓ

Use the bread recipe in ANNEXURE C and answer the questions that follow.

2.2.1 How many grams of active yeast must Mary-Ann use for the bread recipe? (2)

2.2.2 Calculate the maximum temperature of the water in degree celsius that Mary-Ann has to use to make the dough.

Use the following formula:
°C = (°F – 32) ÷ 1,8

Write your final answer to the nearest temperature in degree celsius. (3)

2.2.3 Mary-Ann stated that the loaf pans that she is using is 22,86 cm × 12,7 cm. Show with the necessary calculations whether Mary-Ann is using the correct loaf pans.

Use the following formula:
1 inch = 2,54 cm (3)

2.2.4 Mary-Ann stated that it will take her less than $2\frac{1}{2}$ hours to make the 2 loaves of bread if she only considers the minimum times as indicated in the recipe.

Note:
The time Mary-Ann refers to in her statement includes mixing, rising, panning and baking.

Show, with the necessary calculations, whether Mary-Ann's statement is valid or not. (5)
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QUESTION 3

3.1 A young couple is going to get married. They have chosen a hall for their ceremony and decided on how the seating plan should look like for the reception. The seating plan is represented in ANNEXURE D.

Refer to the seating plan in ANNEXURE D and answer the questions below.

3.1.1 How many people do the couple intend to invite? (3)

3.1.2 Why are there no chairs arranged on the shorter sides of the tables? (2)

3.1.3 A person seated at Table 6 close to the MAIN TABLE wants to put her gift on the GIFT TABLE. Describe a possible route this person can take without walking over the DANCE FLOOR. (2)

3.1.4 Tables 1, 2 and 3 are reserved for the bridal couple, parents of the couple, 1 bridesmaid and 1 best man. Calculate the probability that, if you are invited as a guest, you will be sitting at a table with an even number. Write your final answer to three decimal places. (3)

3.1.5 The caretaker of the hall stated that the area of the dance floor is $\frac{1}{3}$ of the floor area of the hall. The length of the floor is 15,5 m and the width is 9 m. Calculate the area of the dance floor.

You may use the formula:

Area = length × width (3)

3.2 The bridal couple have to spend a lot of money for their entire wedding reception. The reception will be held from 18:00 to 23:30.

Their expenses are as follows:

Hiring of the venue: R3 500,00
Draping and décor: R4 750,00
Disc Jockey (DJ): R250 per hour or part thereof
Catering: R200 per person for the first 100 guests and R100 for every person above 100

Note:
The Disc Jockey will also be included for the catering cost.

The bridal couple stated it will cost R30 000 for the entire reception. Show, with the necessary calculations, whether the statement is valid or not. (6)

[19]

QUESTION 4

4.1 ANNEXURE E shows a map of the Free State in which the Golden Gate National Park is located. Mr. Will, from Edenburg, and his brother from Kroonstad want to visit the Golden Gate National Park during the July holiday. Mr. Will and his brother decide to leave at 10:00 on 10 July 2017.

Refer to the map in ANNEXURE E and answer the following questions.

Mr. Will stated that he and his brother will meet each other at the same time at Marquard. Verify, with an explanation whether this statement is valid or not. (3)

4.2 A weather forecast for Golden Gate National Park from 10 July 2017 to 19 July 2017 is given in ANNEXURE F.

Refer to the weather forecast in ANNEXURE F and answer the following questions.

Determine the following:

4.2.1 Mean for the maximum temperatures (3)

4.2.2 Median for the minimum temperatures (3)

4.2.3 The modal value of the temperatures over the whole period (2)

4.3 Calculate the probability that rain can be expected on one of the days over the period. (2)

[13]

TOTAL: 75

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NOVEMBER 2017

MATHEMATICAL LITERACY P2 ADDENDUM



This ADDENDUM consists of 7 pages with 6 ANNEXURES.

ANNEXURE A

QUESTION 1.1

Account Statement of Mr. Buqa for April 2017

Exclusive Clothing Store

17 Mayfair Road, Sea View, Port Elizabeth
Tel. 041 123 4567 Cell. 089 987 6543

Business hours: Monday to Friday
8:30 am. to 5:00 pm.
Saturday: 8:00 am. to 12:00 pm.

STATEMENT

Store Account Holder: **Mr. Y. Buqa** **29 Sea Street** **Beach View** **Port Elizabeth**	**Date**	01 April 2017
	Statement period	01 March 2017 – 31 March 2017
	Page	1 of 1
	Account Number	3658190

SUMMARY

MONEY OWING ON PREVIOUS PURCHASES: R1 215,80
Interest on outstanding amounts is calculated at 31% per annum.

	This month	Last month
Payment received:	R450,00	R320,00
Total purchases for the month:	R975,00	R215,50
Total refunds:	R135,50	R0,00

TOTAL OWING: R1 636,71
MINIMUM PAYMENT REQUIRED: R327,34

DETAILED DESCRIPTION OF PURCHASES

Date	Description	Code	Debit	Credit
01/04/2017	Interest on money owing		**A**	
06/03/2017	Payment received	----------		R450,00
10/03/2017	Men's jacket	J145875	R476,00	
15/03/2017	Ladies dress	L552632	R135,50	
19/03/2017	Kiddies shoes	C398710	R99,50	
23/03/2017	Refund on returned item	----------		R135,50
26/03/2017	Ladies accessories	L764318	R77,50	
27/03/2017	Kiddies underwear	C444654	R129,50	
29/03/2017	Men's sleepshirt	M265498	R57,00	

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ANNEXURE B

QUESTION 2.1

Absa Gold Current Account	**2016 Fees**	**R500 transaction in 2016**	**2017 Fees**	**R500 transaction in 2017**
Withdrawal (Own Bank)	R3,95 + R1,30 per R100	R10,45	R3,95 + R1,35 per R100	R10,70
Withdrawal (Other ATMs)	R9,95 + R1,30 per R100	R16,45	R9,95 + R1,35 per R100	R16,70
Deposit (ATM)	R3,00 + R1,30 per R100	**A**	R3,95 + R1,35 per R100	R10,70
Debit order (internal)	Free		Free	
Debit order (external)	R16,00		R16,95	
Account fee (PAYT)	R40,00		R42,00	

[Adapted source: www.absa.co.za]

ANNEXURE C

QUESTION 2.2

Basic Homemade Bread Recipe



Ingredients

- 1 package ($\frac{1}{4}$ ounce) active dry yeast
- $2\frac{1}{4}$ cups warm water (110 °F to 115 °F)
- 3 tablespoons sugar
- 1 tablespoon salt
- 2 tablespoons canola oil
- $6\frac{1}{4}$ to $6\frac{3}{4}$ cups all-purpose flour

Directions

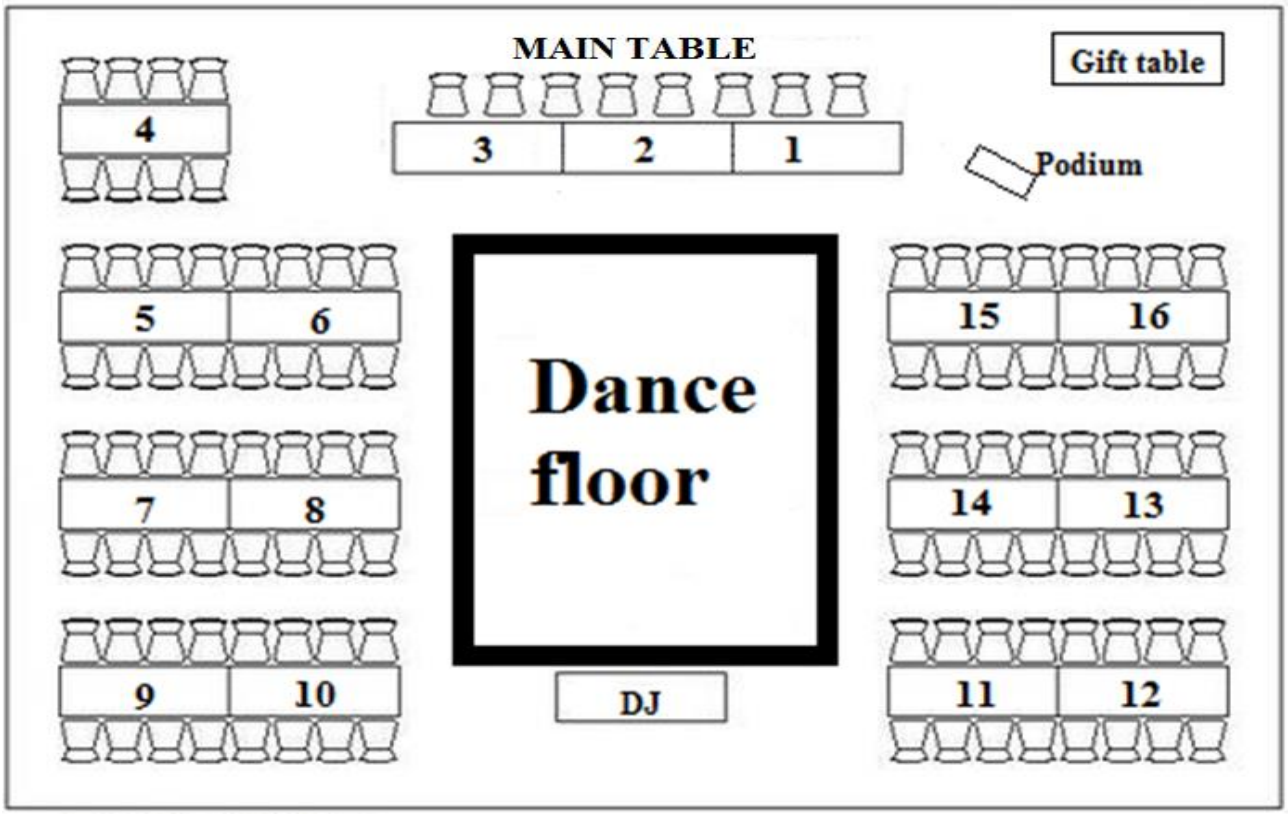
1. In a large bowl, dissolve yeast in warm water. Add the sugar, salt, oil and 3 cups flour. Beat until smooth. Stir in enough remaining flour, $\frac{1}{2}$ cup at a time, to form a soft dough.
2. Turn onto a floured surface; knead until smooth and elastic, about 8–10 minutes. Place in a greased bowl. Cover and let rise in a warm place until doubled, about $1\frac{1}{2}$ hours.
3. Punch dough down. Turn onto a lightly floured surface; divide dough in half. Shape each into a loaf. Place in two greased 9-inch x 5-inch loaf pans. Cover and let rise until doubled, about 30–45 minutes.
4. Bake at 375 F for 30–35 minutes or until golden brown and bread sounds hollow when tapped. Remove from pans to wire racks to cool. Yield: 2 loaves (16 slices each)

[Source: http://www.tasteofhome.com/recipes/basic-homemade-bread]

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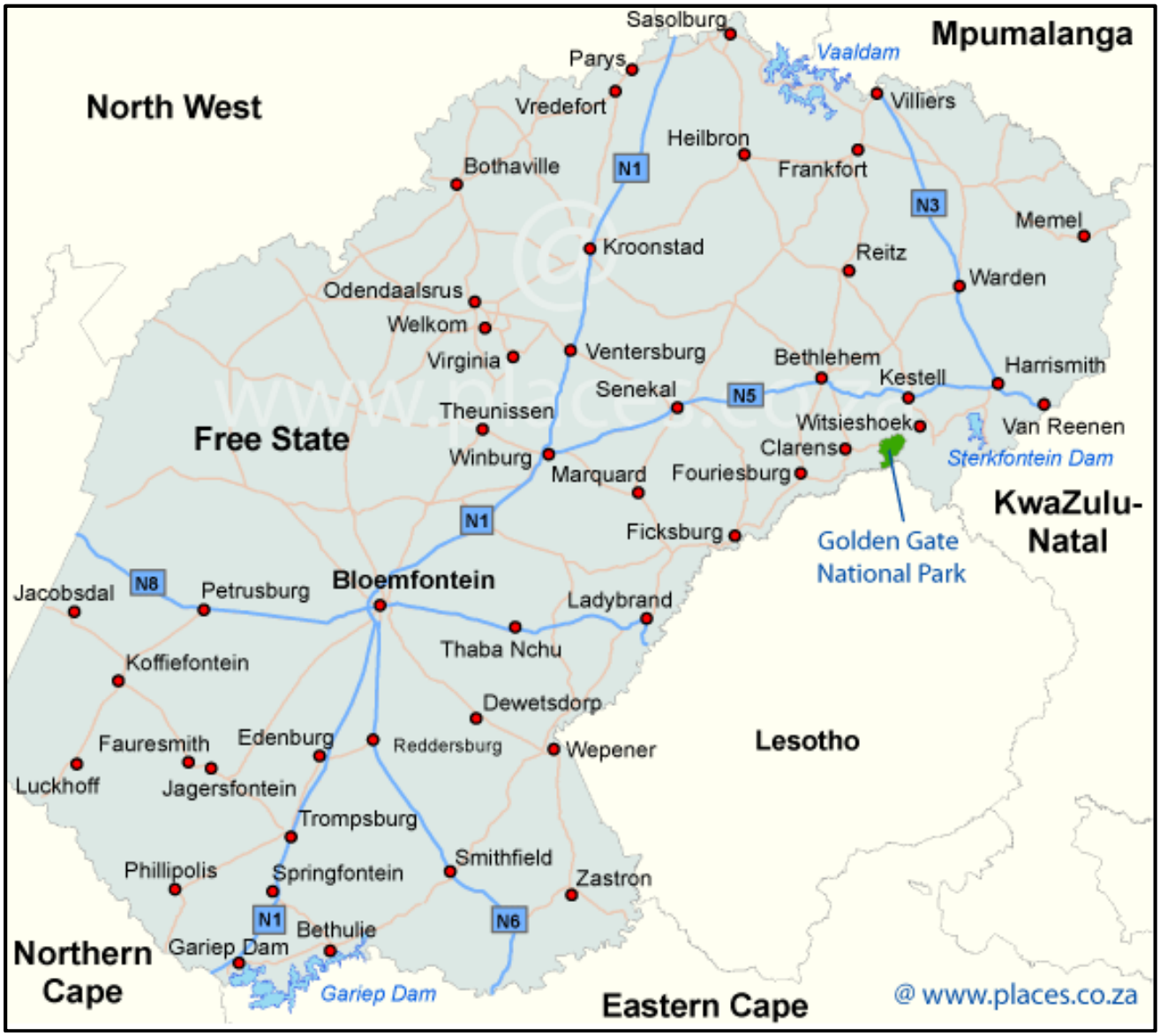
ANNEXURE D

QUESTION 3.1



ANNEXURE E

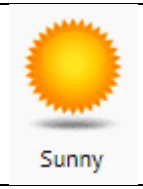
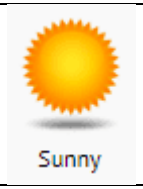
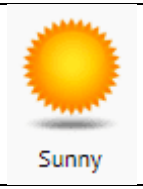
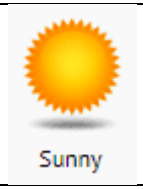
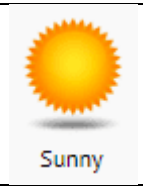
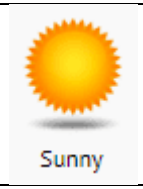
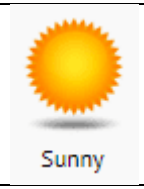
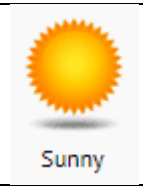
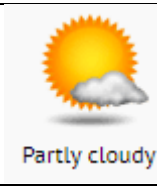
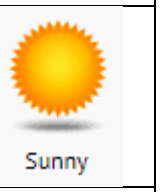
QUESTION 4.1



ANNEXURE F

QUESTION 4.2

The minimum and maximum temperatures for Golden Gate National Park from 10 to 19 July 2017

10 July	11 July	12 July	13 July	14 July	15 July	16 July	17 July	18 July	19 July
 Sunny	 Sunny	 Sunny	 Sunny	 Sunny	 Sunny	 Sunny	 Sunny	 Partly cloudy	 Sunny
MAXIMUM TEMPERATURES									
16 °C	17 °C	17 °C	17 °C	16 °C	15 °C	13 °C	14 °C	13 °C	14 °C
MINIMUM TEMPERATURES									
1 °C	1 °C	1 °C	1 °C	1 °C	−4 °C	−1 °C	−1 °C	−2 °C	0 °C

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